$$\varepsilon_2 = -M_{21}\frac{di_1}{dt}$$

$$\text{(or) } M_{21} = \frac{-\varepsilon_2}{di_1/dt}$$

The negative sign in the above equation shows that the mutually induced emf always opposes the change in current $i_1$ with respect to time. If $\frac{di_1}{dt} = 1\text{A s}^{-1}$, then $M_{21} = -\varepsilon_2$. **Mutual inductance $M_{21}$ is also defined as the opposing emf induced in the coil 2 when the rate of change of current through the coil 1 is $1\text{As}^{-1}$.**

Similarly, if an electric current $i_2$ through coil 2 changes with time, then emf $\varepsilon_1$ is induced in coil 1. Therefore,

$$M_{12} = \frac{N_1\Phi_{12}}{i_2} \quad \text{and} \quad M_{12} = \frac{-\varepsilon_1}{di_2/dt}$$

where $M_{12}$ is the mutual inductance of the coil 1 with respect to coil 2. It can be shown that for a given pair of coils, the mutual inductance is same. i.e.,

$$M_{21} = M_{12} = M$$

In general, the mutual induction between two coils depends on size, shape, the number of turns of the coils, their relative orientation and permeability of the medium.

**Unit of mutual-inductance**

The unit of mutual inductance is also henry (H).

If $i_1 = 1\text{A}$ and $N_2\Phi_{21} = 1$ Wb turns, then $M = 1\text{H}$.

Therefore, **the mutual inductance between two neighbouring coils is said to be one henry if a current of 1A in one coil produces unit flux linkage in neighbouring coil.**

If $\frac{di_1}{dt} = 1\text{As}^{-1}$ and $\varepsilon_2 = -1\text{ V}$, then $M = 1\text{H}$.

Therefore, **the mutual inductance between two neighbouring coils is one henry if a current changing at the rate of $1\text{A s}^{-1}$ in one coil induces an opposing emf of 1V in neighbouring coil.**

## 4.3.4 Mutual inductance between two long co-axial solenoids

Consider two long co-axial solenoids of same length $l$. The length of these solenoids is large when compared to their radii so that the magnetic field produced inside the solenoids is uniform and the fringing effect at the ends may be ignored. Let $A_1$ and $A_2$ be the area of cross section of the solenoids with $A_1$ being greater than $A_2$ as shown in Figure 4.22. The turn density of these solenoids are $n_1$ and $n_2$ respectively.

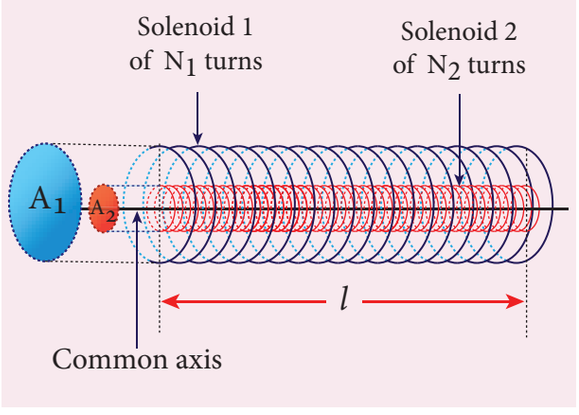



**Figure 4.22** Mutual inductance of two long co-axial solenoids

Let $i_1$ be the current flowing through solenoid 1, then the magnetic field produced inside it is

$$B_1 = \mu_\circ\, n_1\, i_1$$



UNIT-4(XII-Physics_Vol-1).indd 216 17-02-2020 13:18:15